Highlights

Research on the Denominator-Weighted Shapiro Cyclic Sum and Efficient Solution Algorithms

Anqi Wang

- Establishes key properties and characteristic results of the denominator-weighted Shapiro cyclic sum.
- Extends the Bushell–Ando structural characterization of minimizers to the denominator-weighted case.
- Reduces the minimization problem to solving finitely many polynomial equation systems.
- An efficient RBFGS algorithm on the unit sphere is proposed to minimize the Shapiro cyclic sum.

Research on the Denominator-Weighted Shapiro Cyclic Sum and Efficient Solution Algorithms*

Anqi Wang[a,1]

[a]School of Mathematical Sciences, Tongji University, Shanghai, 200092, China

ARTICLE INFO

Keywords:
Shapiro inequality
Shapiro cyclic sum
Riemannian BFGS algorithm
Minimizer analysis

ABSTRACT

This paper investigates the extremal properties and solution strategies of the **denominator-weighted Shapiro cyclic sum**. First, the fundamental properties of the **Shapiro cyclic sum** are derived, the structural characteristics and piecewise patterns of the minimizer of this cyclic sum are explored, and a general method for deriving the minimum of this cyclic sum is presented. To address the issue of extremely complex algebraic computations in high-dimensional cases, this paper transforms the constrained optimization problem into a Riemannian manifold optimization problem on the unit sphere through variable substitution, and proposes an efficient **Riemannian BFGS (RBFGS) algorithm** equipped with an oscillatory solution detection and restart mechanism. Numerical experiments demonstrate that the proposed algorithm exhibits good convergence and efficiency in verifying counterexamples to the classical Shapiro inequality and solving for the minimum of its denominator-weighted form.

1. Introduction

Cyclic inequality is a classical inequality with an elegant structure and a wide range of applications, see[12]. In 1954, Shapiro proposed the following famous cyclic inequality: for $n(n \geq 3)$ nonnegative real numbers $x_1, x_2, \ldots, x_n$,

$$\sum_{i=1}^{n} \frac{x_i}{x_{i+1} + x_{i+2}} \geq \frac{n}{2}, \tag{1.1}$$

where $x_{n+1} = x_1$ and $x_{n+2} = x_2$.

This problem underwent exploration for over thirty years until it was completely resolved by Troesch in 1989: the inequality holds when n is even and $4 \leq n \leq 12$, or when n is odd and $3 \leq n \leq 23$; counterexamples exist when n is even and $n \geq 14$, or when n is odd and $n \geq 25$, see[14][15]. Subsequently, scholars such as Bushell, Ando, and others further investigated the conditions for equality when the inequality holds, see[2][3][1].

To broaden the theoretical depth and application scope, attention has turned to various generalizations of the Shapiro inequality. See, e.g., [5][10][13]. Nguyen and Le [8] systematically studied the following denominator-weighted Shapiro inequality:

$$\sum_{i=1}^{n} \frac{x_i}{px_{i+1} + qx_{i+2}} \geq \frac{n}{p+q}, \tag{1.2}$$

where $p, q \geq 0$ and $p + q > 0$. Clearly, both sides of the denominator-weighted Shapiro inequality are homogeneous with respect to $p + q$, so one may assume $p + q = 1$ without loss of generality. In particular, when $p = q = 1/2$, (1.2) reduces to the classical Shapiro inequality (1.1). Nguyen and Le proved that (1.2) always holds for $n = 3$; for $n = 4$, (1.2) holds if and only if $p \geq q$, i.e., under $p + q = 1$ the necessary and sufficient condition is $\frac{1}{2} \leq p \leq 1$; for $n = 5$, a sufficient condition for the inequality to hold is $\frac{5-\sqrt{5}}{10} \leq p \leq \frac{5+\sqrt{5}}{10}$; when n is even and $0 \leq p < \frac{1}{2}$, (1.2) never holds.

*This work was supported by the Tongji University Student Innovation Training Program (SITP) under No. X2025634.

**Corresponding author
wang_anqi@tongji.edu.cn (A. Wang)
1

In Section 3 of this paper, we will analyze further properties of the denominator-weighted Shapiro inequality, and based on Ando's analysis of the structure of minimizers of the classical Shapiro cyclic sum, systematically extend his method to the denominator-weighted case, summarizing a general approach for finding the minimum point and minimum value of the Shapiro cyclic sum for a given integer n and weight p. To overcome the difficulty of analytical solution for relatively large n, Section 4 further proposes a BFGS algorithm based on Riemannian manifolds, providing theoretical characterization and numerical tools for the study of the minimum of the denominator-weighted Shapiro cyclic sum and the Shapiro inequality problem.

2. Preliminaries

2.1. Known Results of Shapiro Cyclic Sum and Shapiro Inequality in the Denominator-Weighted Form

For an n-dimensional vector $x = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n$ and weight p, denote the denominator-weighted Shapiro cyclic sum by

$$F(x,p) = \sum_{i=1}^{n} \frac{x_i}{px_{i+1} + qx_{i+2}}. \tag{2.1}$$

Then the denominator-weighted Shapiro inequality is

$$F(x,p) = \sum_{i=1}^{n} \frac{x_i}{px_{i+1} + qx_{i+2}} \geq n, \tag{2.2}$$

where $x = (x_1, x_2, \cdots x_n)$, $x_i \geq 0$, $p + q = 1$, $0 \leq p \leq 1$, $x_{n+1} = x_1$, $x_{n+2} = x_2$.

In particular, when $p = \frac{1}{2}$, (2.1) and (2.2) reduce to the classical Shapiro cyclic sum and inequality, respectively, i.e.,

$$F\left(x, \frac{1}{2}\right) = \sum_{i=1}^{n} \frac{x_i}{\frac{1}{2}x_{i+1} + \frac{1}{2}x_{i+2}} \geq n.$$

Note that by continuity, when the denominators are nonzero, there is no difference in studying whether the inequality holds on the domain $x_i \geq 0$ or $x_i > 0$. To facilitate the study of the minimum of the cyclic sum, we set $x_i \geq 0$ in the following.

Theorem 2.1. *For the classical Shapiro inequality, the following hold:*

1. *If n is odd and $3 \leq n \leq 23$, then*

$$F\left(x, \frac{1}{2}\right) = \sum_{i=1}^{n} \frac{x_i}{\frac{1}{2}x_{i+1} + \frac{1}{2}x_{i+2}} \geq n.$$

   Furthermore, equality $F\left(x, \frac{1}{2}\right) = n$ holds if and only if $x_1 = x_2 = \cdots = x_n$.

2. *If n is even and $4 \leq n \leq 12$, then*

$$F\left(x, \frac{1}{2}\right) = \sum_{i=1}^{n} \frac{x_i}{\frac{1}{2}x_{i+1} + \frac{1}{2}x_{i+2}} \geq n.$$

   Furthermore, equality holds if and only if x is of oscillatory type, i.e., there exist $a > 0, b > 0$ such that $(x_1, \cdots, x_n) = (a, b, a, b, \cdots, a, b)$.

3. *If n is even and $n \geq 14$, or n is odd and $n \geq 25$, then there exist $x_1 > 0, \ldots, x_n > 0$ such that $F\left(x, \frac{1}{2}\right) < n$.*

Theorem 2.2 (Nguyen, Le, 2009)**.** *1. When $n = 3$, for any $0 \le p \le 1$, $F(x, p) \ge 3$ always holds.*

2. *When $n = 4$, the inequality $F(x, p) \ge 4$ always holds if and only if $\frac{1}{2} \le p \le 1$.*

3. *When $n = 5$, a sufficient condition for the inequality $F(x, p) \ge 5$ to always hold is $\frac{5-\sqrt{5}}{10} \le p \le \frac{5+\sqrt{5}}{10}$.*

4. *If $n \ge 4$ is even and $0 \le p < \frac{1}{2}$, then there exists $x = (x_1, \dots, x_n) > 0$ such that $F(x, p) < n$, i.e., the inequality does not hold.*

2.2. BFGS Algorithm

There are many tools in numerical computation for optimization problems, such as rank-one update formula, BFGS formula, DFP formula, etc., among which the BFGS formula is the most widely used, see [4].

Theorem 2.3 (BFGS formula for optimization problems)**.** *The BFGS formula for constructing the approximation matrix H_k to $\nabla^2 f(x^{(k)})^{-1}$ is given by:*

$$H_{k+1} = \left(I - \frac{s^{(k)} y^{(k)T}}{y^{(k)T} s^{(k)}} \right) H_k \left(I - \frac{y^{(k)} s^{(k)T}}{y^{(k)T} s^{(k)}} \right) + \frac{s^{(k)} s^{(k)T}}{y^{(k)T} s^{(k)}},$$

or equivalently:

$$H_{k+1} = H_k + \frac{s^{(k)} s^{(k)T}}{y^{(k)T} s^{(k)}} + \frac{y^{(k)T} H_k y^{(k)} s^{(k)} s^{(k)T}}{(y^{(k)T} s^{(k)})^2} - \frac{s^{(k)} y^{(k)T} H_k + H_k y^{(k)} s^{(k)T}}{y^{(k)T} s^{(k)}}.$$

The BFGS algorithm for the unconstrained optimization problem $\min_{x \in \mathbb{R}^n} f(x)$ ($f \in C^1$) proceeds as follows:

1. Initialization: Choose an initial point $x^{(0)} \in \mathbb{R}^n$, set $H_0 = I_n$.
2. For $k = 0, 1, 2, \dots$, perform:
   (a) Compute the gradient $g^{(k)} = \nabla f(x^{(k)})$. If $\|g^{(k)}\| < \varepsilon$ (where ε is a preset tolerance) or the maximum number of iterations is exceeded, output $x^{(k)}$ and terminate.
   (b) Compute the search direction $d^{(k)} = -H_k g^{(k)}$.
   (c) Determine the step size λ_k via a line search satisfying the Wolfe conditions.
   (d) Update the point: $x^{(k+1)} = x^{(k)} + \lambda_k d^{(k)}$.
   (e) Compute the difference vectors:
   $$s^{(k)} = x^{(k+1)} - x^{(k)}, \quad y^{(k)} = \nabla f(x^{(k+1)}) - \nabla f(x^{(k)}).$$
   (f) Update the inverse Hessian approximation using the BFGS formula:
   $$H_{k+1} = \left(I - \frac{s^{(k)} y^{(k)T}}{y^{(k)T} s^{(k)}} \right) H_k \left(I - \frac{y^{(k)} s^{(k)T}}{y^{(k)T} s^{(k)}} \right) + \frac{s^{(k)} s^{(k)T}}{y^{(k)T} s^{(k)}}.$$

3. Minimum of the Denominator-Weighted Shapiro Cyclic Sum

3.1. Notation and Basic Properties

For the denominator-weighted Shapiro cyclic sum

$$F(x, p) = \sum_{i=1}^{n} \frac{x_i}{p x_{i+1} + q x_{i+2}},$$

where $x = (x_1, x_2, \cdots x_n)$, $x_i \ge 0$, $p + q = 1$, $0 \le p \le 1$, $x_{n+1} = x_1$, $x_{n+2} = x_2$. Denote

$$\partial_i F(x, p) = \frac{\partial}{\partial x_i} F(x, p) = \frac{1}{p x_{i+1} + q x_{i+2}} - \frac{q x_{i-2}}{(p x_{i-1} + q x_i)^2} - \frac{p x_{i-1}}{(p x_i + q x_{i+1})^2},$$

$$\partial_p F(x,p) = \frac{\partial}{\partial p} F(x,p) = \sum_{i=1}^{n} \frac{x_i(x_{i+2} - x_{i+1})}{(px_{i+1} + qx_{i+2})^2},$$

$$\begin{aligned}
\overline{K}_n &:= \{(x_1, \cdots, x_n) \in \mathbb{R}^n \mid x_1 \geq 0, \ldots, x_n \geq 0\},\\
K_n^\circ &:= \{(x_1, \cdots, x_n) \in \mathbb{R}^n \mid x_1 > 0, \ldots, x_n > 0\},\\
K_n^\bullet &:= \{(x_1, \cdots, x_n) \in \overline{K}_n \mid (x_1, \ldots, x_n) \notin K_n^\circ, (x_i, x_{i+1}) \neq (0,0), \forall i = 1, 2, \ldots, n\},\\
K_n &:= K_n^\circ \cup K_n^\bullet,\\
P(n,p) &:= \inf_{x \in K_n} F(x,p).
\end{aligned}$$

By the AM-GM inequality, we have

$$P(n,0) = \inf_{x \in K_n} F(x,0) = \inf_{x \in K_n} \sum_{i=1}^{n} \frac{x_i}{x_{i+2}} = n, \quad P(n,1) = \inf_{x \in K_n} F(x,1) = \inf_{x \in K_n} \sum_{i=1}^{n} \frac{x_i}{x_{i+1}} = n.$$

Furthermore, taking $x = (1, 1, \cdots, 1)$, we have $F(x,p) = n$, so $P(n,p) \leq n$.

Theorem 3.1 (Existence of the minimum)**.** *For any integer $n \geq 3$ and $0 < p < 1$, there exists $x \in K_n$ such that $P(n,p) = F(x,p)$.*

Proof. Note that $F(x,p)$ is homogeneous of degree zero: for any $t > 0$, $F(tx,p) = F(x,p)$. Therefore, we consider the domain as the set $\{x \in K_n \mid \sum_{i=1}^n x_i = 1\} = S \cap K_n$, where $S = \{x \in \mathbb{R}^n \mid x_i \geq 0, \sum_{i=1}^n x_i = 1\}$, which is compact. he proof is then completed by this compactness. □

Theorem 3.2. *1.* $P(n+2,p) \leq P(n,p) + \begin{cases} \dfrac{1}{1-p}, & \text{if } p > \dfrac{1}{2},\\ 2, & \text{if } p \leq \dfrac{1}{2}. \end{cases}$

2. For odd $n \geq 3$, $P(n+1,p) \leq P(n,p) + 1$.

Proof. (1) Suppose $P(n,p) = F(x^*,p)$ with $x^* = (x_1, x_2, \cdots, x_n) \in K_n$. Considering $x' = (x_1, x_2, \cdots, x_n, x_1, x_2) \in K_{n+2}$, the proof becomes quite simple.

(2) Suppose $P(n,p) = F(x^*,p)$ with $x^* = (x_1, x_2, \cdots, x_n) \in K_n$. Consider $x^r = (x_1, x_2, \cdots, x_{r-1}, x_r, x_r, x_{r+1}, \cdots, x_n) \in K_{n+1}$. Then

$$F(x^r,p) - F(x^*,p) - 1 = \frac{x_{r-1}}{px_r + qx_r} + \frac{x_r}{px_r + qx_{r+1}} - \frac{x_{r-1}}{px_r + x_{r+1}} - 1 = \frac{q(x_r - x_{r-1})(x_r - x_{r+1})}{px_r^2 + qx_r x_{r+1}}$$

If there exists r such that $(x_r - x_{r-1})(x_r - x_{r+1}) \leq 0$, then we obtain

$$P(n+1,p) \leq F(x^r,p) \leq F(x^*,p) + 1 = P(n,p) + 1.$$

Otherwise, if for all r, $(x_r - x_{r-1})(x_r - x_{r+1}) > 0$, then

$$\prod_{r=1}^{n} (x_r - x_{r+1})^2 = \prod_{r=1}^{n} (x_{r-1} - x_r)(x_r - x_{r+1}) < 0,$$

a contradiction. □

Theorem 3.3. *$P(n,p)$ is a continuous function of $p \in (0,1)$.*

Proof. Consider the domain as the set $\{x \in K_n \mid \sum_{i=1}^n x_i = 1\} = S \cap K_n$, where $S = \{x \in \mathbb{R}^n \mid x_i \geq 0, \sum_i x_i = 1\}$, which is compact.

Fix n and p_0. There exists $x^* \in S \cap K_n$ such that $F(x^*, p_0) = P(n, p_0)$. For this x^*, there exists $\epsilon_0 > 0$ such that for all i, x_i and x_{i+1} are not both $< \epsilon_0$. Denote $T_n^i(\epsilon) = \{x \in \mathbb{R}^n \mid x_i < \epsilon, x_{i+1} < \epsilon\}$. Then for any $0 < \epsilon \leq \epsilon_0$, $x^* \in A(\epsilon) = S \setminus \cup_{i=1}^n T_n^i(\epsilon) \subseteq S \cap K_n$, where $A(\epsilon) = S \setminus \cup_{i=1}^n T_n^i(\epsilon)$ is always compact.

If there exist $0 < \epsilon \le \epsilon_0$ and $\delta > 0$ such that for all $p \in (p_0 - \delta, p_0 + \delta) \cap (0, 1)$, $F(x, p)$ attains its minimum at some $x^p \in A(\epsilon) = S \setminus \cup_{i=1}^{n} T_n^i(\epsilon)$, then by Berge's Maximum Theorem, considering $F(x, p)$ as a continuous function on a compact set $A(\epsilon) \times [p_0 - \delta/2, p_0 + \delta/2]$, where $A(\epsilon)$ does not change with p within a small range, the value function $P(n, p) = \min_{x \in A(\epsilon)} F(x, p)$ is a continuous function of the parameter p. □

Lemma 3.4. *There exist $\epsilon > 0$ and $\delta > 0$ such that for all $p \in (p_0 - \delta, p_0 + \delta) \cap (0, 1)$, $F(x, p)$ attains its minimum at some $x^p \in A = S \setminus \bigcup_{i=1}^{n} T_n^i(\epsilon)$.*

Proof. Suppose otherwise. For any sequences $\epsilon_k \to 0$ and $\delta_k \to 0$ ($k = 1, 2, \ldots$), there exist a sequence of parameters p_k and a corresponding sequence of minimizers $x^{(k)}$ satisfying:

- $|p_k - p_0| < \delta_k$, i.e., $p_k \to p_0$.
- $x^{(k)}$ is a minimizer of $F(x, p_k)$ on $S \cap K_n$, and $x^{(k)} \in S \cap (\cup_{i=1}^{n} T_n^i(\epsilon_k))$. That is, for each k, there exists some index j_k such that $x_{j_k}^{(k)} < \epsilon_k$ and $x_{j_k+1}^{(k)} < \epsilon_k$.

Since S is compact in $\mathbb{R}^n$, the sequence $\{x^{(k)}\}$ must have a convergent subsequence. Without loss of generality, denote this convergent subsequence by $\{x^{(k)}\}$ and let its limit be $x^{**} \in S$, i.e., $x^{(k)} \to x^{**} \in S$ as $k \to +\infty$.

For each k, there exists an index j_k such that two consecutive components are less than ϵ_k. Since there are only n indices (finitely many), there must exist some index j such that $j_k = j$ infinitely often in the subsequence. Consider this subsequence and assume $x_j^{(k)} < \epsilon_k$ and $x_{j+1}^{(k)} < \epsilon_k$ for all k. Taking the limit $k \to +\infty$, by sign preservation we have

$$x_j^{**} = 0 \quad \text{and} \quad x_{j+1}^{**} = 0.$$

If $x_{j-1}^{**} > 0$, then the term $\frac{x_{j-1}^k}{p_k x_j^k + q_k x_{i+1}^k}$ in $\{F(x_k, p_k)\}_k$ tends to $+\infty$, a contradiction.
If $x_{j-1}^{**} = 0$, proceeding inductively backward, and considering $x^{**} \in S$, we again obtain a contradiction. □

Definition 3.5. *We say that $x = (x_1, \cdots, x_n) \in K_n$ and $y = (y_1, \cdots, y_n) \in K_n$ have the same pattern if $x_i = 0 \Leftrightarrow y_i = 0$ for all $i = 1, \ldots, n$.*

Let $x = (x_1, \cdots, x_n) \in K_n^{\bullet}$. If $x_i = 0$, $x_{i+1} \neq 0, \ldots, x_{i+l} \neq 0$, and $x_{i+l+1} = 0$, where $i, l \ge 1$, then $(x_{i+1}, \cdots, x_{i+l})$ is called a segment of x, and l is called the length of this segment.

For a segment $s := (x_i, \cdots, x_j)$ of x, we denote

$$\mathrm{S}(s) := \sum_{k=i}^{j-1} \frac{x_k}{p x_{k+1} + q x_{k+2}}, \quad \mathit{Head}(s) := x_i, \quad \mathit{Tail}(s) := x_j.$$

Here, if the length of s is 1, then $\mathrm{S}(s) = 0$.

Let $s_1, \ldots, s_r$ be all segments of x. Let l_k be the length of s_k. The tuple $(l_1, \cdots, l_r)$ is called the pattern of x. We have

$$F(x, p) = \sum_{k=1}^{r} \mathrm{S}(s_k) + \sum_{k=1}^{r} \frac{\mathrm{Tail}(s_{k-1})}{q\,\mathrm{Head}(s_k)},$$

where we adopt the convention $s_{k+r} = s_k$ for $k \in \mathbb{Z}$, and the indices of segments are periodic with period r.

Theorem 3.6. *Assume that $P(n, p) = \min_{x \in K_n^{\bullet}} F(x, p) = F(x^*, p)$ is attained at $x^* \in K_n^{\bullet}$. Let $s_1, \ldots, s_r$ be all segments of x^* (arranged in order), and let l_k be the length of s_k. Then the following conclusions hold.*

1. $\frac{\mathrm{Tail}(s_1)}{\mathrm{Head}(s_2)} = \frac{\mathrm{Tail}(s_2)}{\mathrm{Head}(s_3)} = \cdots = \frac{\mathrm{Tail}(s_{r-1})}{\mathrm{Head}(s_r)} = \frac{\mathrm{Tail}(s_r)}{\mathrm{Head}(s_1)}$.
2. *Assume $x^* = (s_1, 0, s_2, 0, \cdots, s_r, 0)$, and let σ be a permutation of $\{1, 2, \ldots, r\}$. Then there exist positive real numbers $t_1, t_2, \ldots, t_r > 0$ such that the vector $x' = (t_1 s_{\sigma(1)}, 0, t_2 s_{\sigma(2)}, 0, \cdots, t_r s_{\sigma(r)}, 0)$ satisfies $F(x', p) = F(x^*, p)$.*

The proof follows the same lines as Theorem 2.4 in [1].

Corollary 3.7. *For a minimizer x^* for fixed n and p, we may assume without loss of generality that $x^* = (s_1, 0, s_2, 0, \dots, s_r, 0)$ and its pattern $(l_1, \cdots, l_r)$ satisfies $l_1 \geq l_2 \geq \cdots \geq l_r$. We always arrange the pattern of such an x^* in descending order, i.e., the segment lengths are sorted from largest to smallest.*

Definition 3.8. *Suppose $x \in K_n^\bullet$ satisfies the form of conclusion (1) in the above theorem. Denote*

$$U(x) := \frac{\mathrm{Tail}(s_1)}{\mathrm{Head}(s_2)} = \frac{\mathrm{Tail}(s_2)}{\mathrm{Head}(s_3)} = \cdots = \frac{\mathrm{Tail}(s_{r-1})}{\mathrm{Head}(s_r)} = \frac{\mathrm{Tail}(s_r)}{\mathrm{Head}(s_1)}.$$

Then we have $F(x,p) = \frac{r}{q}U(x) + \sum_{k=1}^{r} \mathrm{S}(s_k)$.

3.2. Structural Characterization of Minimizers

For fixed n, p and $x = (x_1, \cdots, x_n) \in K_n$, denote:

$$\begin{aligned}
a_i(x) &:= \frac{x_i}{px_{i+1} + qx_{i+2}},\\
B(x) &:= (px_2 + qx_3, px_3 + qx_4, \cdots, px_n + qx_1, px_1 + qx_2),\\
R(x) &:= \left(\frac{1}{x_n}, \frac{1}{x_{n-1}}, \frac{1}{x_{n-2}}, \dots, \frac{1}{x_1}\right),\\
T(x) &= \left(\frac{x_n}{(px_1 + qx_2)^2}, \dots, \frac{x_{n+1-i}}{(px_{n+2-i} + qx_{n+3-i})^2}, \dots, \frac{x_1}{(px_2 + qx_3)^2}\right).
\end{aligned}$$

Denote the i-th element of $B(x)$ by $B(x)_i = px_{i+1} + qx_{i+2}$, and similarly define $R(x)_i$ and $T(x)_i$.

Lemma 3.9. *1.* $\partial_i F(x,p) = (R(B(x)))_{n+1-i} - (B(T(x)))_{n+1-i}$, $\quad \sum_{i=1}^{n} x_i \partial_i F(x,p) = 0.$

2. $(T^2(x))_i = \frac{x_i}{(1 - B(x)_i \cdot \partial_i F(x,p))^2}$.

3. $F(T(x),p) - F(x,p) = \sum_{i=1}^{n} \frac{x_i (\partial_i F(x,p))^2}{B(T(x))_{n+1-i}}$.

4. $F(x,p) + F(y,p) = F(x+y,p) + F(T(x)+T(y),p) - \sum_{i=1}^{n} \frac{(T(x)+T(y))_{n+1-i}(\partial_i F(x,p) + \partial_i F(y,p))}{(R(B(x))+R(B(y)))_{n+1-i} \cdot B(T(x)+T(y))_{n+1-i}}$.

Proof. (1) and (2) follow from direct calculation.

(3)

$$F(T(x),p) - F(x,p) = \sum_{i=1}^{n} \frac{(T(x))_i}{B(T(x))_i} - \sum_{i=1}^{n} \frac{x_i}{B(x)_i} = \sum_{i=1}^{n} \left(\frac{(T(x))_i}{B(T(x))_i} - \frac{x_i}{B(x)_i}\right).$$

Change the index $i \mapsto n+1-i$, and use $T(x)_{n+1-i} = \dfrac{x_i}{(B(x)_i)^2}$ and from (1) we have $B(T(x))_{n+1-i} = \dfrac{1}{B(x)_i} - \partial_i F(x,p)$. Then

$$\begin{aligned}
F(T(x),p) - F(x,p) &= \sum_{i=1}^{n} \left(\frac{x_i/(B(x)_i)^2}{\frac{1}{B(x)_i} - \partial_i F(x,p)} - \frac{x_i}{B(x)_i}\right) && = \sum_{i=1}^{n} x_i \partial_i F(x,p) \left(\frac{1}{1 - B(x)_i \partial_i F(x,p)} - 1\right)\\
&= \sum_{i=1}^{n} \frac{x_i}{B(x)_i} \left(\frac{1}{1 - B(x)_i \partial_i F(x,p)} - 1\right) && = \sum_{i=1}^{n} \frac{x_i (\partial_i F(x,p))^2 B(x)_i}{1 - B(x)_i \partial_i F(x,p)}\\
&= \sum_{i=1}^{n} \frac{x_i \partial_i F(x,p)}{1 - B(x)_i \partial_i F(x,p)} && = \sum_{i=1}^{n} \frac{x_i (\partial_i F(x,p))^2}{B(T(x))_{n+1-i}}.\\
&= \sum_{i=1}^{n} \frac{x_i \partial_i F(x,p)}{1 - B(x)_i \partial_i F(x,p)} - \sum_{i=1}^{n} x_i \partial_i F(x,p)
\end{aligned}$$

(4) Fix an index i, and let $a = x_i,\ b = B(x)_i,\ c = y_i,\ d = B(y)_i$. Then

$$a_i(x) = \frac{a}{b}, \quad a_i(y) = \frac{c}{d}, \quad a_i(x+y) = \frac{a+c}{b+d}.$$

Direct calculation yields the identity

$$\frac{a}{b} + \frac{c}{d} = \frac{a+c}{b+d} + \frac{a/b^2 + c/d^2}{1/b + 1/d}.$$

By definition,

$$a/b^2 = T(x)_{n+1-i}, \quad c/d^2 = T(y)_{n+1-i}, \quad 1/b = (R(B(x)))_{n+1-i}, \quad 1/d = (R(B(y)))_{n+1-i}.$$

Thus,

$$\frac{x_i}{B(x)_i} + \frac{y_i}{B(y)_i} = \frac{a}{b} + \frac{c}{d} = \frac{a+c}{b+d} + \frac{a/b^2 + c/d^2}{1/b + 1/d} = \frac{(x+y)_i}{B(x+y)_i} + \frac{(T(x)+T(y))_{n+1-i}}{(R(B(x)) + R(B(y)))_{n+1-i}}. \tag{3.1}$$

Summing (3.1) over i, we obtain

$$F(x,p) + F(y,p) = F(x+y,p) + \sum_{i=1}^{n} \frac{(T(x)+T(y))_{n+1-i}}{(R(B(x)) + R(B(y)))_{n+1-i}}. \tag{3.2}$$

On the other hand, from equation (1) we have

$$\partial_i F(x,p) = (R(B(x)))_{n+1-i} - B(T(x))_{n+1-i}, \quad \partial_i F(y,p) = (R(B(y)))_{n+1-i} - B(T(y))_{n+1-i}.$$

Since B is a linear map,

$$B(T(x)+T(y))_{n+1-i} = B(T(x))_{n+1-i} + B(T(y))_{n+1-i}.$$

Hence,

$$(R(B(x)) + R(B(y)))_{n+1-i} - B(T(x)+T(y))_{n+1-i} = \partial_i F(x,p) + \partial_i F(y,p).$$

Therefore,

$$\begin{aligned} &\frac{(T(x)+T(y))_{n+1-i}}{B(T(x)+T(y))_{n+1-i}} - \frac{(T(x)+T(y))_{n+1-i}}{(R(B(x)) + R(B(y)))_{n+1-i}} \\ &= \frac{(T(x)+T(y))_{n+1-i}\left((R(B(x)) + R(B(y)))_{n+1-i} - B(T(x)+T(y))_{n+1-i}\right)}{(R(B(x)) + R(B(y)))_{n+1-i} B(T(x)+T(y))_{n+1-i}} \\ &= \frac{(T(x)+T(y))_{n+1-i}(\partial_i F(x,p) + \partial_i F(y,p))}{(R(B(x)) + R(B(y)))_{n+1-i} B(T(x)+T(y))_{n+1-i}}. \end{aligned} \tag{3.3}$$

Summing both sides of (3.3) over i and noting that

$$\sum_{i=1}^{n} \frac{(T(x)+T(y))_{n+1-i}}{B(T(x)+T(y))_{n+1-i}} = \sum_{i=1}^{n} \frac{(T(x)+T(y))_i}{B(T(x)+T(y))_i} = F(T(x)+T(y),p),$$

we obtain

$$F(T(x)+T(y),p) - \sum_{i=1}^{n} \frac{(T(x)+T(y))_{n+1-i}}{(R(B(x)) + R(B(y)))_{n+1-i}} = \sum_{i=1}^{n} \frac{(T(x)+T(y))_{n+1-i}(\partial_i F(x,p) + \partial_i F(y,p))}{(R(B(x)) + R(B(y)))_{n+1-i} B(T(x)+T(y))_{n+1-i}}. \tag{3.4}$$

Substituting (3.4) into (3.2) yields

$$F(x,p) + F(y,p) = F(x+y,p) + F(T(x)+T(y),p) - \sum_{i=1}^{n} \frac{(T(x)+T(y))_{n+1-i}(\partial_i F(x,p) + \partial_i F(y,p))}{(R(B(x)) + R(B(y)))_{n+1-i} B(T(x)+T(y))_{n+1-i}}.$$

This is exactly equation (4). □

Theorem 3.10. *1. For $x \in K_n$, we have $F(T(x), p) \geq F(x, p)$. Moreover, if $F(T(x), p) = F(x, p)$, then $T^2(x) = x$.*

2. If $P(n, p) = \min_{x \in K_n} F(x, p) = F(x^*, p)$ is attained at $x^* \in K_n^\bullet$, then the following hold:

$$T^2(x^*) = x^*, \quad F(T(x^*), p) = F(x^*, p).$$

Proof. (1) It follows directly from Lemma 3.9(3) that $F(T(x), p) \geq F(x, p)$. Assume $F(T(x), p) = F(x, p)$. Then by Lemma 3.9(3), we have $x_i(\partial_i F(x, p))^2 = 0$ for all $i = 1, \ldots, n$. Hence $x_i = 0$ or $\partial_i F(x, p) = 0$. By Lemma 3.9(2), we obtain $(T^2(x))_i = x_i$.

(2) If $\min_{x \in K_n} F(x, p) = F(x^*, p)$ is attained at $x^* \in K_n^\bullet$, then for all $i = 1, 2, \ldots, n$, either $x_i = 0$ or $\partial_i F(x^*, p) = 0$. By Lemma 3.9(2), $T^2(x^*)$ coincides with x^*; by Lemma 3.9(3), $F(T(x^*), p) = F(x^*, p)$. □

Lemma 3.11. *Let a, b, c, d, e be nonnegative real numbers and p, q be real numbers. If for any $\lambda \geq 0$,*

$$p\frac{1 + \lambda a}{(1 + \lambda c)^2} + q\frac{1 + \lambda b}{(1 + \lambda d)^2} = \frac{1}{1 + \lambda e}, \tag{3.5}$$

then the following conclusions hold:

1. *If $p = 0$, then $q = 1$ and $b = d = e$.*
2. *If $q = 0$, then $p = 1$ and $a = c = e$.*
3. *If $p \neq 0$ and $q \neq 0$, then $c = d = e$.*

Note that both sides of this equality are rational fractions in λ with nonzero denominators; the proof follows by clearing denominators and comparing coefficients of the resulting polynomials.

Theorem 3.12. *1. Suppose there exist $a, b \in K_n^\bullet$ having the same pattern (i.e., the positions of zero components are identical) and satisfying*

$$\min_{x \in K_n} F(x, p) = F(a, p) = F(b, p).$$

Then there exists a real number $\mu > 0$ such that $a = \mu b$.

2. Suppose $P(n, p) = \min_{x \in K_n} F(x, p) = F(x^*, p)$ is attained at some $x^* \in K_n^\circ$. Then $F(x^*, p) = n$, and x^* is either a constant sequence $(a, a, \ldots, a)$ $(a > 0)$, or, when n is even and $p = \frac{1}{2}$, an oscillatory sequence $(a, b, a, b, \ldots, a, b)$ $(a, b > 0)$.

Proof. (1) Since a and b have the same pattern, the positions of their zero components are identical. Let $I = \{i : a_i = 0\}$; then $b_i = 0$ for $i \in I$. By the minimality condition, for $i \notin I$ we have $\partial_i F(a, p) = 0$ and $\partial_i F(b, p) = 0$. Consider any $\lambda > 0$ and set $y = \lambda b$. Then $F(y, p) = F(b, p) = F(a, p)$. For $i \notin I$, $\partial_i F(y, p) = 0 = \partial_i F(b, p)$; for $i \in I$, $a_i = y_i = 0$. Hence for all i,

$$(T(a) + T(\lambda b))_{n+1-i}(\partial_i F(a, p) + \partial_i F(\lambda b, p)) = 0.$$

Applying Lemma 3.9(4) with $x = a$ and $y = \lambda b$, we obtain

$$F(a, p) + F(\lambda b, p) = F(a + \lambda b, p) + F(T(a) + T(\lambda b), p).$$

Since $\min_{x \in K_n} F(x, p) = F(a, p) = F(b, p) = F(\lambda b, p)$, we have

$$\min_{x \in K_n} F(x, p) = F(a + \lambda b, p) = F(T(a) + T(\lambda b), p), \quad \forall \lambda > 0.$$

Thus for all $i \notin I$,

$$0 = \partial_i F(a + \lambda b, p) = \frac{1}{B(a + \lambda b)_i} - \frac{q(a + \lambda b)_{i-2}}{[B(a + \lambda b)_{i-2}]^2} - \frac{p(a + \lambda b)_{i-1}}{[B(a + \lambda b)_{i-1}]^2}, \tag{3.6}$$

Introduce parameters

$$\alpha = \frac{b_{i-2}}{a_{i-2}}, \quad \beta = \frac{b_{i-1}}{a_{i-1}}, \quad \gamma = \frac{B(b)_{i-2}}{B(a)_{i-2}}, \quad \delta = \frac{B(b)_{i-1}}{B(a)_{i-1}}, \quad \varepsilon = \frac{B(b)_i}{B(a)_i}, \quad r = \frac{qa_{i-2}B(a)_i}{[B(a)_{i-2}]^2}, \quad s = \frac{pa_{i-1}B(a)_i}{[B(a)_{i-1}]^2}.$$

Expressing the components of $(a + \lambda b)$ and the corresponding B values in terms of these parameters, we have

$$\alpha, \beta \geq 0, \quad \gamma, \delta, \varepsilon > 0, \quad r, s \geq 0.$$

Substituting into (3.6) and simplifying yields an identity in λ:

$$r\frac{1 + \lambda\alpha}{(1 + \lambda\gamma)^2} + s\frac{1 + \lambda\beta}{(1 + \lambda\delta)^2} = \frac{1}{1 + \lambda\varepsilon}, \quad \forall \lambda \geq 0. \tag{3.7}$$

Here, if $i - 2 \in I$, then $r = 0$; if $i - 1 \in I$, then $s = 0$; if $i - 2, i - 1 \notin I$, then $r \neq 0, s \neq 0$. The three cases of equation (3.7) correspond to the three cases in Lemma 3.11 of the form (3.5).

If $i - 2 \in I$, then $r = 0$, $s = 1$, and $\beta = \delta = \varepsilon$:

$$\frac{b_{i-1}}{a_{i-1}} = \frac{B(b)_{i-1}}{B(a)_{i-1}} = \frac{B(b)_i}{B(a)_i}. \tag{3.8}$$

If $i - 1 \in I$, then $s = 0$, $r = 1$, and $\alpha = \gamma = \varepsilon$:

$$\frac{b_{i-2}}{a_{i-2}} = \frac{B(b)_{i-2}}{B(a)_{i-2}} = \frac{B(b)_i}{B(a)_i}. \tag{3.9}$$

If $i - 2, i - 1 \notin I$, then $r \neq 0, s \neq 0$, and $\gamma = \delta = \varepsilon$:

$$\frac{B(b)_{i-2}}{B(a)_{i-2}} = \frac{B(b)_{i-1}}{B(a)_{i-1}} = \frac{B(b)_i}{B(a)_i}. \tag{3.10}$$

Write the segments of a as $(s_1, 0, s_2, 0, \cdots, 0, s_r, 0)$, and similarly b as $(t_1, 0, t_2, 0, \cdots, 0, t_r, 0)$. The corresponding segments of a and b have the same lengths, arranged in decreasing order.

We analyze the different cases for $i \notin I$. **If** $i \notin I$ **lies in a segment of length** 1, then consider (3.9):

$$\frac{b_{i-2}}{a_{i-2}} = \frac{b_i}{a_i} = \frac{b_{i+2}}{a_{i+2}} = \frac{B(b)_i}{B(a)_i} = \frac{B(b)_{i-1}}{B(a)_{i-1}} = \frac{B(b)_{i-2}}{B(a)_{i-2}} = \frac{B(b)_{i-3}}{B(a)_{i-3}}.$$

If $i \notin I$ **lies in a segment of length** ≥ 2 **and is the first element of that segment**, then consider (3.9):

$$\frac{b_{i-2}}{a_{i-2}} = \frac{b_i}{a_i} = \frac{B(b)_i}{B(a)_i} = \frac{B(b)_{i-2}}{B(a)_{i-2}} = \frac{B(b)_{i-3}}{B(a)_{i-3}}.$$

If $i \notin I$ **lies in a segment of length** ≥ 2 **and is the second element of that segment**, then consider (3.8):

$$\frac{b_{i-1}}{a_{i-1}} = \frac{B(b)_{i-1}}{B(a)_{i-1}} = \frac{B(b)_i}{B(a)_i}.$$

If $i \notin I$ **lies in a segment of length** ≥ 3 **and is the third or later element of that segment**, then consider (3.10):

$$\frac{B(b)_{i-2}}{B(a)_{i-2}} = \frac{B(b)_{i-1}}{B(a)_{i-1}} = \frac{B(b)_i}{B(a)_i}.$$

In summary, for all $i \notin I$, we have $\frac{B(b)_i}{B(a)_i} \equiv \frac{1}{\mu}$ constant, i.e.,

$$pa_{i+1} + qa_{i+2} = \mu(pb_{i+1} + qb_{i+2}), \quad \forall i \notin I.$$

For $i \in I$, then $i+1 \notin I$. Applying (3.9) with index $i+1$, we obtain

$$\frac{b_{i-1}}{a_{i-1}} = \frac{b_{i+1}}{a_{i+1}} = \frac{1}{\mu}.$$

Thus the ratios of the first elements and of the last elements of corresponding segments of b and a are both $\frac{1}{\mu}$. Consequently,

$$\forall i = 1, 2, \dots, n, \quad a_i = \mu b_i,$$

i.e., $a = \mu b$.

(2) If $\min\limits_{x \in K_n} F(x,p) = F(x^*,p)$ is attained at some $x^* \in K_n^\circ$, then this x^* has only one segment. Write $x^* = (x_1, x_2, \cdots, x_n)$ and consider $x' = (x_2, x_3, \cdots, x_n, x_1) \in K_n^\circ$. Then $F(x^*,p) = F(x',p) = \min\limits_{x \in K_n} F(x,p)$.

Similar to the proof of (1), only the third case in Lemma 3.11 is needed. For all $i = 1, 2, \dots, n$,

$$\begin{aligned} \frac{B(x')_i}{B(x^*)_i} = \frac{B(x^*)_{i+1}}{B(x^*)_i} \equiv \frac{1}{\mu} = 1, \\ px_{i+2} + qx_{i+3} = px_{i+1} + qx_{i+2}, \\ qx_{i+3} + (p-q)x_{i+2} - px_{i+1} = 0. \end{aligned}$$

The matrix form of this system is:

$$\begin{pmatrix} 0 & -p & p-q & q & 0 & \cdots & 0 \\ 0 & 0 & -p & p-q & q & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & -p & p-q & q \\ q & 0 & \cdots & 0 & 0 & -p & p-q \\ p-q & q & \cdots & 0 & 0 & 0 & -p \\ -p & p-q & q & 0 & \cdots & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Consider the characteristic equation of the recurrence relation:

$$q\lambda^2 + (p-q)\lambda - p = 0.$$

The characteristic roots are:

$$\lambda_1 = 1, \quad \lambda_2 = -\frac{p}{q}.$$

Hence the general solution takes the form:

$$x_i = A + B\left(-\frac{p}{q}\right)^i, \quad i = 1, 2, \dots, n.$$

Substituting the periodicity condition $x_{i+n} = x_i$, we obtain:

$$B\left(-\frac{p}{q}\right)^i \left[\left(-\frac{p}{q}\right)^n - 1\right] = 0, \quad \forall i.$$

- If $B = 0$, then $x_i = A$, i.e., a constant sequence, which always holds.
- If $B \neq 0$, then $\left(-\frac{p}{q}\right)^n = 1$. Since $-\frac{p}{q} \leq 0$, it must be -1, yielding $p = q = \frac{1}{2}$ and n even.

Furthermore, consider the cases $q = 0$, $p = 1$ or $p = 0$, $q = 1$; in each case only constant solutions exist.

In summary, the solution space of this system is as follows:

Case 1: $p = q = \frac{1}{2}$ **and** n **even**. The dimension of the solution space is 2, with a basis:

$$\boldsymbol{\xi}_1 = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}^T, \quad \boldsymbol{\xi}_2 = \begin{pmatrix} 1 & -1 & 1 & -1 & \cdots & 1 & -1 \end{pmatrix}^T,$$

and the general solution is $\mathbf{x} = A\boldsymbol{\xi}_1 + B\boldsymbol{\xi}_2$, where A, B are constants. In this case, $x \in K_n^\circ$ is an oscillatory sequence.

Case 2: Otherwise. The dimension of the solution space is 1, with a basis:

$$\boldsymbol{\xi} = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}^T,$$

and the general solution is $\mathbf{x} = C\boldsymbol{\xi}$, where C is an constant. In this case, $x \in K_n^\circ$ is a constant sequence. □

Corollary 3.13. *Assume that* $P(n,p) = \min\limits_{x \in K_n} F(x,p) = F(x^*,p)$ *is attained at* $x^* \in K_n^\bullet$. *Let s and t be two segments of x^* of equal length. Then there exists a constant $c > 0$ such that $s = ct$.*

Proof. By Theorem 3.6(2), let σ be a permutation that swaps only s and t. Then there exists x' having the same pattern as x^* and satisfying $\min\limits_{x \in K_n} F(x,p) = F(x',p) = F(x^*,p)$. By Theorem 3.12, there exists a real number $\mu > 0$ such that $x' = \mu x^*$, hence there exists $c > 0$ such that $s = ct$. □

Theorem 3.14. *Let* $P(n,p) = \min\limits_{x \in K_n} F(x,p) = F(x^*,p)$ *be attained at* $x^* \in K_n^\bullet$. *Let* $s = (x_1, x_2, \ldots, x_l)$ *be a segment of x^* of length $l \geq 2$, and denote* $U = U(x) = \frac{\mathrm{Tail}(s_1)}{\mathrm{Head}(s_2)} = \frac{\mathrm{Tail}(s_2)}{\mathrm{Head}(s_3)} = \cdots = \frac{\mathrm{Tail}(s_{r-1})}{\mathrm{Head}(s_r)} = \frac{\mathrm{Tail}(s_r)}{\mathrm{Head}(s_1)} = \frac{x_l}{x_{l+2}}$. *Then there exists a real number $\mu > 0$ such that*

$$\left(\frac{x_l}{(qx_{l+2})^2}, \frac{x_{l-1}}{(px_l)^2}, \frac{x_{l-2}}{(px_{l-1}+qx_l)^2}, \cdots, \frac{x_2}{(px_3+qx_4)^2}, \frac{x_1}{(px_2+qx_3)^2} \right) = \mu(x_1, x_2, \cdots, x_l).$$

Equivalently,

$$\left(\frac{U^2}{q^2 x_l}, \frac{x_{l-1}}{p^2 x_l^2}, \frac{x_{l-2}}{(px_{l-1}+qx_l)^2}, \cdots, \frac{x_2}{(px_3+qx_4)^2}, \frac{x_1}{(px_2+qx_3)^2} \right) = \mu(x_1, x_2, \cdots, x_l).$$

Proof. Without loss of generality, assume $x = (s, 0, \cdots)$, i.e., $x_1, \ldots, x_l$ are nonzero, $x_{l+1} = 0$, and x_{l+2} is the first element of the next segment. By Theorem 3.10(2), $F(T(x^*),p) = F(x^*,p) = \min\limits_{x \in K_n} F(x,p)$. According to the relationships between the components of $T(x)$ and x, there exists a permutation σ of the segments of $T(x^*)$ such that each segment of x^* is aligned with the corresponding segment of $\sigma(T(x^*))$ at the same positions. Then x^* and $\sigma(T(x^*))$ have the same pattern. By Theorem 3.12(1), there exists a real number $\mu > 0$ such that $\sigma(T(x^*)) = \mu x^*$. Considering $s = (x_1, x_2, \cdots, x_l)$ and its corresponding segment $T(s) = \left(\frac{x_l}{(px_{l+1}+qx_{l+2})^2}, \frac{x_{l-1}}{(px_l+qx_{l+1})^2}, \ldots, \frac{x_1}{(px_2+qx_3)^2} \right)$, after simplification we obtain: there exists a real number $\mu > 0$ such that

$$\left(\frac{x_l}{(qx_{l+2})^2}, \frac{x_{l-1}}{(px_l)^2}, \frac{x_{l-2}}{(px_{l-1}+qx_l)^2}, \cdots, \frac{x_2}{(px_3+qx_4)^2}, \frac{x_1}{(px_2+qx_3)^2} \right) = \mu(x_1, x_2, \cdots, x_l),$$

$$\left(\frac{U^2}{q^2 x_l}, \frac{x_{l-1}}{p^2 x_l^2}, \frac{x_{l-2}}{(px_{l-1}+qx_l)^2}, \cdots, \frac{x_2}{(px_3+qx_4)^2}, \frac{x_1}{(px_2+qx_3)^2} \right) = \mu(x_1, x_2, \cdots, x_l).$$

□

Now define a sequence of multivariate polynomials $P_k(z_1, \cdots, z_k)$ $(k \geq -1)$ as follows:

$$P_{-1} := 0, \quad P_0 := 1, \quad P_1(z_1) := pz_1, \quad P_k(z_1, \cdots, z_k) := z_1\big(pP_{k-1}(z_2, \cdots, z_k) + qP_{k-2}(z_3, \cdots, z_k)\big) \quad (k \geq 2).$$

Lemma 3.15. *The following conclusions hold:*

1. *For $k \geq 1$,*

$$P_k(z_1, \cdots, z_k) = pz_k P_{k-1}(z_1, \cdots, z_{k-1}) + qz_{k-1} P_{k-2}(z_1, \cdots, z_{k-2}).$$

2. *For $1 \leq j < k$,*

$$P_k(z_1, \cdots, z_k) = P_j(z_1, \cdots, z_j)\, P_{k-j}(z_{j+1}, \cdots, z_k) + qz_j P_{j-1}(z_1, \cdots, z_{j-1})\, P_{k-j-1}(z_{j+2}, \cdots, z_k).$$

Proof. Consider the k-order tridiagonal matrix

$$M_k = \begin{pmatrix} pz_1 & -1 & 0 & \cdots & 0 & 0 & 0 \\ qz_1 & pz_2 & -1 & \cdots & 0 & 0 & 0 \\ 0 & qz_2 & pz_3 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & pz_{k-2} & -1 & 0 \\ 0 & 0 & 0 & \cdots & qz_{k-2} & pz_{k-1} & -1 \\ 0 & 0 & 0 & \cdots & 0 & qz_{k-1} & pz_k \end{pmatrix}.$$

Expanding $\det(M_k)$ along the first row and then along the last row, it is easy to obtain

$$\det(M_k) = pz_1 P_{k-1}(z_2, \cdots, z_k) + qz_1 P_{k-2}(z_3, \cdots, z_k) = P_k(z_1, \cdots, z_k),$$

$$\det(M_k) = pz_k P_{k-1}(z_1, \cdots, z_{k-1}) + qz_{k-1} P_{k-2}(z_1, \cdots, z_{k-2}) = P_k(z_1, \cdots, z_k).$$

Next, perform a Laplace expansion on the first j rows of M_k. Due to the tridiagonal structure of M_k, only two choices of columns yield nonzero contributions: (1) Select the first j columns. Then the minor is $\det(M_j) = P_j(z_1, \cdots, z_j)$, the cofactor is $\det(M_{k-j}(z_{j+1}, \cdots, z_k)) = P_{k-j}(z_{j+1}, \cdots, z_k)$, and the sign is positive. (2) Select the first $j-1$ columns and the $(j+1)$-th column. Then the minor is $-P_{j-1}(z_1, \cdots, z_{j-1})$, the cofactor is $qz_j P_{k-j-1}(z_{j+2}, \cdots, z_k)$, and the sign is negative. This term yields $qz_j P_{j-1}(z_1, \cdots, z_{j-1}) P_{k-j-1}(z_{j+2}, \cdots, z_k)$. Any other selection leads to a zero minor or zero cofactor. Therefore,

$$P_k(z_1, \cdots, z_k) = P_j(z_1, \cdots, z_j) P_{k-j}(z_{j+1}, \cdots, z_k) + qz_j P_{j-1}(z_1, \cdots, z_{j-1}) P_{k-j-1}(z_{j+2}, \cdots, z_k).$$

□

In the following, denote

$$K_n^{\triangle} := \{(x_1, \cdots, x_n) \in K_n^{\bullet} \mid x_{n-1} = 1,\ x_n = 0\}, \qquad y_i := \frac{x_i}{px_{i+1} + qx_{i+2}} = a_i(x), \quad i = 1, \ldots, n.$$

Then for $x = (x_1, \cdots, x_n) \in K_n^{\triangle}$, we have $y_n = 0$, $y_{n-1} = \frac{x_{n-1}}{px_n + qx_1} = \frac{1}{qx_1}$, $y_{n-2} = \frac{x_{n-2}}{px_{n-1} + qx_n} = \frac{x_{n-2}}{p}, \cdots$. Conversely, given $x \in K_n^{\triangle}$, the corresponding $(y_1, y_2, \cdots, y_n)$ can be obtained by solving the system of equations $y_i(px_{i+1} + qx_{i+2}) - x_i = 0$ (for $i = 1, \ldots, n-2$) for x.

Lemma 3.16. *Let $x = (x_1, \cdots, x_n) \in K_n^{\triangle}$ and $\Phi(x) = (y_1, \cdots, y_n)$. Assume that $x_i\, \partial_i F(x, p) = 0$ for all $i = 1, \ldots, n$. Then:*

1. *For $i = 1, \ldots, n-2$,*

$$\frac{y_i}{x_i} = q\, y_1^2\, P_{i-1}(y_1, \cdots, y_{i-1}).$$

2. *For $i = 2, \ldots, n-2$,*

$$y_1 \geq y_i.$$

Proof. Let $p_i := P_i(y_1, \dots, y_i)$. Then (1) can be written as $y_i = q\, y_1^2\, p_{i-1}\, x_i$.

(1) From $x_i\, \partial_i F(x,p) = 0$ we obtain

$$y_i = \frac{q x_{i-2}}{(p x_{i-1} + q x_i)^2}\, x_i + \frac{p x_{i-1}}{(p x_i + q x_{i+1})^2}\, x_i,$$

i.e.,

$$\frac{y_i}{x_i} = q\, \frac{y_{i-2}^2}{x_{i-2}} + p\, \frac{y_{i-1}^2}{x_{i-1}} \qquad (i = 1, \dots, n), \tag{3.11}$$

with the conventions $x_0 = x_n = 0,\ x_{-1} = x_{n-1} = 1,\ y_0 = 0,\ y_{-1} = 1/(q x_1)$. We now prove:

$$\frac{y_i}{x_i} = q\, y_1^2\, p_{i-1} \qquad (i = 1, \dots, n-2). \tag{3.12}$$

For $i = 1$, from (3.11) we have

$$\frac{y_1}{x_1} = q\, \frac{y_{-1}^2}{x_{-1}} + p\, \frac{y_0^2}{x_0} = q\, \frac{1/(q x_1)^2}{1} = \frac{1}{q x_1^2},$$

$$q x_1 y_1 = 1, \qquad \frac{y_1}{x_1} = q y_1^2 = q y_1^2 p_0.$$

Thus (3.12) holds.

For $i = 2$, from (3.11) and the already obtained $y_1/x_1 = q y_1^2$ we get

$$\frac{y_2}{x_2} = q\, \frac{y_0^2}{x_0} + p\, \frac{y_1^2}{x_1} = pq\, y_1^3 = q y_1^2 p_1.$$

Assume that for all $j < i$ we have $y_j/x_j = q y_1^2 p_{j-1}$. Then for $i \geq 3$,

$$\begin{aligned} \frac{y_i}{x_i} &= q\, \frac{y_{i-2}^2}{x_{i-2}} + p\, \frac{y_{i-1}^2}{x_{i-1}} \\ &= q\, y_{i-2}\, (q y_1^2 p_{i-3}) + p\, y_{i-1}\, (q y_1^2 p_{i-2}) \\ &= q\, y_1^2 (q y_{i-2} p_{i-3} + p y_{i-1} p_{i-2}) \\ &= q\, y_1^2 p_{i-1}. \end{aligned}$$

(2) Consider the system of equations in the unknowns $x_1, x_2, \cdots, x_{i-1}$:

$$\begin{cases} -x_1 + p y_1 x_2 + q y_1 x_3 = 0 \\ -x_2 + p y_2 x_3 + q y_2 x_4 = 0 \\ \qquad \vdots \\ -x_{i-2} + p y_{i-2} x_{i-1} = -q y_{i-2} x_i \\ -x_{i-1} = -p y_{i-1} x_i - q y_{i-1} x_{i+1}. \end{cases}$$

In matrix form,

$$\begin{pmatrix} -1 & p y_1 & q y_1 & 0 & \cdots & 0 \\ 0 & -1 & p y_2 & q y_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & p y_{i-3} & q y_{i-3} \\ 0 & \cdots & 0 & 0 & -1 & p y_{i-2} \\ 0 & \cdots & 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{i-3} \\ x_{i-2} \\ x_{i-1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ -q y_{i-2} x_i \\ -p y_{i-1} x_i - q y_{i-1} x_{i+1} \end{pmatrix}.$$

By Cramer's rule,

$$x_1 = (-1)^{i-1} \det \begin{pmatrix} 0 & py_1 & qy_1 & 0 & \cdots & 0 \\ 0 & -1 & py_2 & qy_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & py_{i-3} & qy_{i-3} \\ -qy_{i-2}x_i & \cdots & 0 & 0 & -1 & py_{i-2} \\ -py_{i-1}x_i - qy_{i-1}x_{i+1} & \cdots & 0 & 0 & 0 & -1 \end{pmatrix}$$

$$= (-1)^{i-1}(-qy_{i-2}x_i)(-1)^{i-1} \det \begin{pmatrix} py_1 & qy_1 & 0 & \cdots & 0 \\ -1 & py_2 & qy_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & -1 & py_{i-3} & qy_{i-3} \\ 0 & \cdots & 0 & 0 & -1 \end{pmatrix}$$

$$+ (-1)^{i-1}(-py_{i-1}x_i - qy_{i-1}x_{i+1})(-1)^{i} \det \begin{pmatrix} py_1 & qy_1 & 0 & \cdots & 0 \\ -1 & py_2 & qy_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \cdots & 0 & -1 & py_{i-3} & qy_{i-3} \\ 0 & \cdots & 0 & -1 & py_{i-2} \end{pmatrix}$$

$$= qy_{i-2}x_ip_{i-3} + (py_{i-1}x_i + qy_{i-1}x_{i+1})p_{i-2}$$
$$= (qy_{i-2}p_{i-3} + py_{i-1}p_{i-2})x_i + qy_{i-1}p_{i-2}x_{i+1}$$
$$= p_{i-1}x_i + qy_{i-1}p_{i-2}x_{i+1}.$$

Multiplying both sides by y_1^2 and using $qx_1y_1 = 1$ yields

$$y_1 = qy_1^2p_{i-1}x_i + q^2y_1^2y_{i-1}p_{i-2}x_{i+1}.$$

From (1) we have $y_i = qy_1^2p_{i-1}x_i$ and $y_{i-1} = qy_1^2p_{i-2}x_{i-1}$. Substituting these into the above gives

$$y_1 - y_i = q^2y_1^2y_{i-1}p_{i-2}x_{i+1} = \frac{qy_{i-1}^2x_{i+1}}{x_{i-1}} \geq 0.$$

□

Theorem 3.17. *If* $\min_{x \in K_n} F(x,p) = F(x^*,p)$ *is attained at* $x^* \in K_n^\bullet$, *then* $U(x^*) \geq q$, *where* $U(x) = \frac{\text{Tail}(s_1)}{\text{Head}(s_2)} = \frac{\text{Tail}(s_2)}{\text{Head}(s_3)} = \cdots = \frac{\text{Tail}(s_{r-1})}{\text{Head}(s_r)} = \frac{\text{Tail}(s_r)}{\text{Head}(s_1)}$.

Proof. Without loss of generality, let $x^* = (x_1, \cdots, x_n) \in K_n^{\triangle}$. By the minimality condition, we have $x_i\, \partial_i F(x,p) = 0$ for all i, and $U(x^*) = \frac{1}{x_1} = qy_1$. By Lemma 3.16, for all $i = 1, 2, \ldots, n$,

$$0 \leq \frac{x_i}{px_{i+1} + qx_{i+2}} = y_i \leq y_1 = \frac{1}{qx_1} = \frac{U(x^*)}{q}.$$

If $U(x^*) < q$, then $x_1 > \frac{1}{q}$, and

$$x_i < px_{i+1} + qx_{i+2},$$

$$\sum_{i=1}^{n} x_i < \sum_{i=1}^{n}(px_{i+1} + qx_{i+2}) = p\sum_{i=2}^{n+1} x_i + q\sum_{i=3}^{n+2} x_i = \sum_{i=1}^{n} x_i,$$

a contradiction. □

3.3. General Procedure for Finding the Minimum

Theorem 3.18. *For all $n \geq 3$, if $\frac{\sqrt{5}-1}{2} \leq p \leq 1$, then $P(n,p) = \min\limits_{x \in K_n} F(x,p) = n$.*

Proof. We claim that if $\frac{\sqrt{5}-1}{2} < p \leq 1$, then $P(n,p) = \min\limits_{x \in K_n} F(x,p) = F(x^*,p)$ is attained at some $x^* \in K_n^\circ$. Then by Theorem 3.12(2), $P(n,p) = F(x^*,p) = n$. Consequently, by Theorem 3.3, for $\frac{\sqrt{5}-1}{2} \leq p \leq 1$, we have $P(n,p) = n$.

If not, then $P(n,p) = \min\limits_{x \in K_n} F(x,p) = F(x^*,p)$ is attained at some $x^* \in K_n^\bullet$. Then x^* contains zeros. Let $s = (x_1, x_2, \cdots, x_l)$ be a segment of x^* and $x_{l+1} = 0$.

Denote $U = U(x) = \frac{x_l}{x_{l+2}}$ and $V = \frac{x_{l-1}+x_l}{x_l} \geq 1$. Consider

$$
\begin{aligned}
0 \leq \partial_{l+1} F(x^*,p) &= \frac{1}{px_{l+2}+qx_{l+3}} - \frac{qx_{l-1}}{(px_l+qx_{l+1})^2} - \frac{px_l}{(px_{l+1}+qx_{l+2})^2} \\
&= \frac{1}{px_{l+2}+qx_{l+3}} - \frac{qx_{l-1}}{(px_l)^2} - \frac{px_l}{(qx_{l+2})^2} \\
&\leq \frac{1}{px_{l+2}} - \frac{q(V-1)}{p^2x_l} - \frac{px_l}{(qx_{l+2})^2} \\
&= \frac{U}{px_l} - \frac{q(V-1)}{p^2x_l} - \frac{pU^2}{q^2x_l} \\
&= \frac{1}{x_l}\left(\frac{U}{p} - \frac{q(V-1)}{p^2} - \frac{pU^2}{q^2}\right).
\end{aligned}
$$

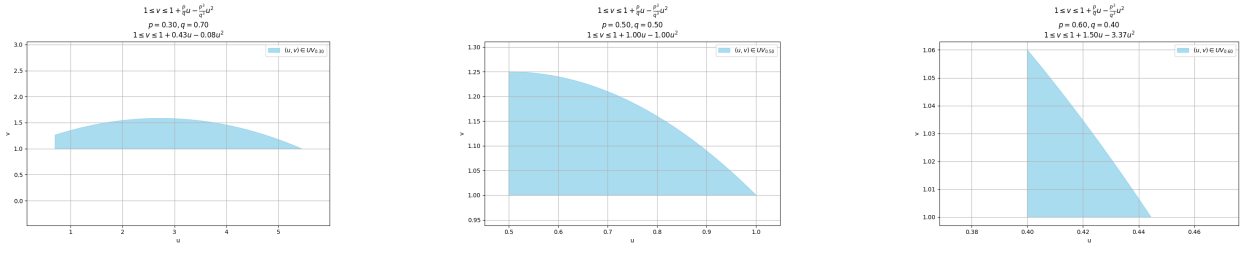
Hence

$$
1 \leq V \leq 1 + \frac{p}{q}U - \frac{p^3}{q^3}U^2 = -\frac{p^3}{q^3}\left(U - \frac{q^2}{2p^2}\right)^2 + 1 + \frac{q^4}{4p^4}.
$$

From $1 \leq 1 + \frac{p}{q}U - \frac{p^3}{q^3}U^2$ we obtain $0 \leq U \leq \frac{q^2}{p^2}$. Moreover, by Theorem 3.17, $U \geq q$. It is easy to verify that when $\frac{\sqrt{5}-1}{2} < p \leq 1$, the two intervals have no intersection. This is a contradiction. □

According to Theorem 3.12(2), the case where $P(n,p) = \min\limits_{x \in K_n} F(x,p) = F(x^*,p)$ is attained at some $x^* \in K_n^\circ$ is already well understood. In the following, we consider the case where $P(n,p) = \min\limits_{x \in K_n} F(x,p) = F(x^*,p)$ is attained at some $x^* \in K_n^\bullet$. Then $U = U(x)$ lies in the interval $\left[q, \frac{q^2}{p^2}\right]$, and the feasible region for (U,V) is $UV_p = \left\{(u,v) \mid q \leq u \leq \frac{q^2}{p^2}, 1 \leq v \leq 1 + \frac{p}{q}u - \frac{p^3}{q^3}u^2\right\}$.

Example 3.19. *Write a Python program* [16] *to plot the feasible region UV_p for a specified value of p.*



(a) $p = 0.3$ (b) $p = 0.5$ (c) $p = 0.6$

Figure 1: The feasible regions for $p = 0.3$, $p = 0.5$ and $p = 0.6$

Let $s = (x_1, x_2, \cdots, x_l)$ be a segment of x^* with $x_{l+1} = 0$, $x_{-1} = x_{n-1} = Ux_1$, $x_0 = x_n = 0$. From the result of Theorem 3.14,

$$\left(\frac{U^2}{q^2 x_l}, \frac{x_{l-1}}{p^2 x_l^2}, \frac{x_{l-2}}{(px_{l-1} + qx_l)^2}, \cdots, \frac{x_2}{(px_3 + qx_4)^2}, \frac{x_1}{(px_2 + qx_3)^2} \right) = \mu(x_1, x_2, \cdots, x_l).$$

$$\frac{1}{\mu} = \frac{q^2 x_1 x_l}{U^2} = \frac{p^2 x_2 x_l^2}{x_{l-1}}$$

$$x_2 = \frac{q^2 x_1 x_{l-1}}{p^2 U^2 x_l} = \frac{q^2(V-1)}{p^2 U^2} x_1.$$

For all $i = 3, 4, \ldots, l+2$, $x_{i-2} \neq 0$, we have $\partial_{i-2} F(x, p) = 0$.

$$\partial_{i-2} F(x, p) = \frac{1}{px_{i-1} + qx_i} - \frac{qx_{i-4}}{(px_{i-3} + qx_{i-2})^2} - \frac{px_{i-3}}{(px_{i-2} + qx_{i-1})^2} = 0$$

$$x_i = \frac{1}{\dfrac{q^2 x_{i-4}}{(px_{i-3} + qx_{i-2})^2} + \dfrac{pqx_{i-3}}{(px_{i-2} + qx_{i-1})^2}} - \frac{p}{q} \cdot x_{i-1}.$$

Let $x_i = f_i^l x_1$ (the superscript l is an index, not an exponent). Then $f_{-1}^l = U$, $f_0^l = 0$, $f_1^l = 1$, $f_2^l = \frac{q^2(V-1)}{p^2 U^2}$, and for $i \geq 3$,

$$f_i^l = \frac{1}{\dfrac{q^2 f_{i-4}^l}{(pf_{i-3}^l + qf_{i-2}^l)^2} + \dfrac{pqf_{i-3}^l}{(pf_{i-2}^l + qf_{i-1}^l)^2}} - \frac{p}{q} f_{i-1}^l.$$

Example 3.20. *Write a Python program* [16] *using the SymPy package to compute f_i^l for specified $i \geq -1$.*

$$f_3^l = \frac{pu - qv + q}{pu^2}.$$

$$f_4^l = \frac{-p^2 u + pqv - pq + q^2}{pqu^2}.$$

$$f_5^l = \frac{p^3 uv - p^3 u + p^2 qu - p^2 qv^2 + 2p^2 qv - p^2 q - 2pq^2 v + 2pq^2}{pq^2 u^2 v - pq^2 u^2 + q^3 u^2}.$$

$f_6^l = (-p^7u^2v^3 + 3p^7u^2v^2 - 3p^7u^2v + p^7u^2 - 3p^6qu^2v^2 + 6p^6qu^2v - 3p^6qu^2 + 2p^6quv^4 - 8p^6quv^3 + 12p^6quv^2 - 8p^6quv + 2p^6qu - 3p^5q^2u^2v + 3p^5q^2u^2 + 7p^5q^2uv^3 - 21p^5q^2uv^2 + 21p^5q^2uv - 7p^5q^2u - p^5q^2v^5 + 5p^5q^2v^4 - 10p^5q^2v^3 + 10p^5q^2v^2 - 5p^5q^2v + p^5q^2 - p^4q^3u^2 + 7p^4q^3uv^2 - 14p^4q^3uv + 7p^4q^3u - 4p^4q^3v^4 + 16p^4q^3v^3 - 24p^4q^3v^2 + 16p^4q^3v - 4p^4q^3 + 2p^3q^4uv - 2p^3q^4u - 4p^3q^4v^3 + 12p^3q^4v^2 - 12p^3q^4v + 4p^3q^4 + pq^6v - pq^6 + q^7)/(p^4q^3u^3v^3 - 3p^4q^3u^3v^2 + 3p^4q^3u^3v - p^4q^3u^3 + 3p^3q^4u^3v^2 - 6p^3q^4u^3v + 3p^3q^4u^3 - p^3q^4u^2v^4 + 4p^3q^4u^2v^3 - 6p^3q^4u^2v^2 + 4p^3q^4u^2v - p^3q^4u^2 + 3p^2q^5u^3v - 3p^2q^5u^3 - 3p^2q^5u^2v^3 + 9p^2q^5u^2v^2 - 9p^2q^5u^2v + 3p^2q^5u^2 + pq^6u^3 - 2pq^6u^2v^2 + 4pq^6u^2v - 2pq^6u^2).$

For $p = 0.3$, compute $f_6^l = (-0.0002187u^2v^3 - 0.0008748u^2v^2 - 0.0011664u^2v - 0.0005184u^2 + 0.0010206uv^4 + 0.0042525uv^3 + 0.0005670uv^2 - 0.0050085uv - 0.0008316u - 0.0011907v^5 - 0.0051597v^4 + 0.0066150v^3 + 0.0230202v^2 - 0.0039984v + 0.0630679)/(0.0027783u^3v^3 + 0.0111132u^3v^2 + 0.0148176u^3v + 0.0065856u^3 - 0.0064827u^2v^4 - 0.0194481u^2v^3 + 0.0266511u^2v^2 + 0.0309729u^2v - 0.0316932u^2)$.

That is, from (U, V) and x_1, all elements of this segment $\{x_i\}_{i=1}^l$ can be determined. If $P(n, p) = \min_{x \in K_n} F(x, p) = F(x^*, p)$ is attained at some $x^* \in K_n^\bullet$, and x^* contains a segment of length l, then there must exist a point (U, V) in the feasible region UV_p determined by p such that for all $i = 1, 2, \cdots, l$, $f_i^l > 0$, and $f_{l+1}^l = 0$.

By Corollary 3.13, if $P(n,p) = \min\limits_{x\in K_n} F(x,p) = F(x^*,p)$ is attained at some $x^* \in K_n^\bullet$, then any two segments s and t of x^* having the same length l must be proportional, thus share the same value of V. For $x \in K_n^\bullet$, we define $R_l(x^*) = \frac{\text{Head}(s)}{\text{Tail}(s)}$. From the above derivation, $R_l(x^*) = \frac{1}{f_l^l}$. For a segment of length 1, $R_1(x^*) = 1$. Note the definitions of U and R. For $x = (s_1, 0, s_2, 0, \cdots, s_k, 0)$ with pattern $(l_1, l_2, \cdots, l_k)$, we have the relation

$$U(x)^k R_{l_1} R_{l_2} \cdots R_{l_k} = 1.$$

In UV_p, consider the set of points (u,v) such that for all $i = 1, 2, \cdots, l$, $f_i^l(u,v) > 0$ and $f_{l+1}^l(u,v) = 0$. Then the projection of this set onto U yields an interval, and $R_l(x^*) = \frac{1}{f_l^l}$ also ranges over some interval on this set.

For given n and p, consider all possible patterns $(l_1, l_2, \cdots, l_k)$ with total length n. Compute the feasible region UV_p and ranges of U and R_l in UV_p corresponding to each l_i. If the minimizer has pattern $(l_1, l_2, \cdots, l_k)$, then the U ranges must have a nonempty intersection, and the equation $U^k R_{l_1} R_{l_2} \cdots R_{l_k} = 1$ must have a solution within that intersection. That is, we need to solve the following problem:

Find $(u, v_1, v_2, \cdots, v_k)$ satisfying:

$$\begin{cases} \forall i = 1, 2, \ldots, k, \\ \qquad (u, v_i) \in UV_p = \left\{ (u,v) \mid q \le u \le \frac{q^2}{p^2}, 1 \le v \le 1 + \frac{p}{q}u - \frac{p^3}{q^3}u^2 \right\} \\ \qquad \forall j = 1, 2, \ldots, l_i, \quad f_j^{l_i}(u, v_i) > 0, \quad f_{l_i+1}^{l_i}(u, v_i) = 0. \\ u^k = f_{l_1}^{l_1}(u, v_1) \cdot f_{l_2}^{l_2}(u, v_2) \cdots f_{l_k}^{l_k}(u, v_k). \end{cases} \tag{Q}$$

In fact, for all $j \ge -1$, the expressions $f_j^{l_i}$ are the same rational functions in (u,v) regardless of l_i; we may omit the superscript. The closure conditions in the above problem can be reduced to a system of polynomial equations.

By Corollary 3.13 and Theorem 3.6(2), we can group segments of the same length together. Suppose the minimizer has pattern

$$(\underbrace{l_1, \cdots, l_1}_{m_1 \text{ times}}, \underbrace{l_2, \cdots, l_2}_{m_2 \text{ times}}, \cdots, \underbrace{l_k, \cdots, l_k}_{m_k \text{ times}}).$$

The above problem can be reformulated as follows.

Find $(u, v_1, \cdots, v_k)$ satisfying:

$$\begin{cases} f_{l_i+1}(u, v_i) = 0 \qquad i = 1, 2, \cdots, k \qquad \text{(i)} \\ u^{m_1+\cdots+m_k} = (f_{l_1}(u, v_1))^{m_1} \cdot (f_{l_2}(u, v_2))^{m_2} \cdots (f_{l_k}(u, v_k))^{m_k} \qquad \text{(k+1)} \\ q \le u \le \frac{q^2}{p^2} \\ \forall\, i = 1, 2, \ldots, k, \quad 1 \le v_i \le 1 + \dfrac{p}{q}u - \dfrac{p^3}{q^3}u^2 \\ \qquad \forall\, j = 1, 2, \ldots, l_i, \quad f_j(u, v_i) > 0. \end{cases} \tag{Q}$$

In fact, this problem often has no solution. Since the total number of possible patterns for a given total length n is finite, we can check them one by one and identify all patterns that admit a solution. If no pattern yields a solution, then $P(n,p) = \min\limits_{x\in K_n} F(x,p) = \min\limits_{x\in K_n^\circ} F(x,p)$, which is the trivial case, and the result follows from Theorem 3.12(2).

If there are several patterns with solutions, note that for each pattern, the solution set lies in the compact set

$$(u, v_1, v_2, \cdots, v_k) \in \left[q, \frac{q^2}{p^2}\right] \times \left[1, 1 + \frac{q^4}{4p^4}\right]^k = \widetilde{UV_p} \subset \mathbb{R}^{k+1}.$$

Then by Definition 3.8,

$$\widetilde{F(u,v)} = F(x,p) = \frac{r}{q}U(x) + \sum_{i=1}^{k} \mathrm{S}(s_i)$$

$$= \frac{r}{q}u + \sum_{i=1}^{k}\sum_{j=1}^{l_i-1} \frac{x_j}{px_{j+1} + qx_{j+2}} \text{ (the } j\text{-th element of the } i\text{-th segment)}$$
$$= \frac{r}{q}u + \sum_{i=1}^{k}\sum_{j=1}^{l_i-1} \frac{f_j(u, v_i)}{pf_{j+1}(u, v_i) + qf_{j+2}(u, v_i)}$$

is a continuous function on this compact set and therefore attains a minimum. In fact, for patterns with solutions, the solution set often consists of only a few isolated points. We can compute these minima, compare the minimal values among the finitely many patterns that admit solutions, and then compare with the trivial case value n in Theorem 3.12(2) to obtain $P(n, p) = \min_{x \in K_n} F(x, p)$.

3.3.1. Procedure for Finding the Minimum of the Denominator-Weighted Shapiro Cyclic Sum

1. For a given positive integer $n \geq 3$ and $0 < p < 1$, list all finitely many possible patterns $(l_1, l_2, \cdots, l_k)$ satisfying $l_1 \geq l_2 \geq \cdots \geq l_k$ and $l_1 + l_2 + \cdots + l_k + k = n$.

2. For each pattern, perform the following:

   (a) Combine segments of the same length and rewrite the pattern in the form:

   $$(\underbrace{l_1, \cdots, l_1}_{m_1 \text{ times}}, \underbrace{l_2, \cdots, l_2}_{m_2 \text{ times}}, \cdots, \underbrace{l_k, \cdots, l_k}_{m_k \text{ times}}).$$

   (b) Solve the system of equations in $(u, v_1, \cdots, v_k)$:

   $$\begin{cases} f_{l_i+1}(u, v_i) = 0 \qquad i = 1, 2, \cdots, k & \text{(i)} \\ u^{\sum_{i=1}^{k} m_i} = \prod_{i=1}^{k} (f_{l_i}(u, v_i))^{m_i} & \text{(k+1)} \\ q \leq u \leq \frac{q^2}{p^2} & \\ \forall\, i = 1, 2, \ldots, k, \quad 1 \leq v_i \leq 1 + \frac{p}{q}u - \frac{p^3}{q^3}u^2 & \\ \qquad \forall\, j = 1, 2, \ldots, l_i, \quad f_j(u, v_i) > 0. & \end{cases} \tag{Q}$$

   (c) If the system has no solution, discard this pattern. If the system has a solution, substitute to obtain the value of $\widetilde{F(u, v)} = \frac{r}{q}u + \sum_{i=1}^{k} m_i \sum_{j=1}^{l_i-1} \frac{f_j(u,v_i)}{pf_{j+1}(u,v_i)+qf_{j+2}(u,v_i)}$.

3. Compare all finitely many $\widetilde{F(u, v)}$ obtained from solvable patterns with the trivial case value n to obtain $P(n, p) = \min_{x \in K_n} F(x, p)$.

Example 3.21. *Find* $P(6, 0.3)$.

Consider all patterns with total length 6: (5), (3, 1), (2, 2), (1, 1, 1).
For pattern (5), problem (Q) is to find (u, v) satisfying:

$$\begin{cases} f_6(u, v) = 0 & (1) \\ u = f_5(u, v) & (2) \\ 0.7 \leq u \leq \frac{49}{9} & \\ 1 \leq v \leq 1 + \frac{3}{7}u - \frac{27}{343}u^2 & \\ \forall j = 1, 2, \ldots, 5, \quad f_j(u, v) > 0. & \end{cases} \tag{Q}$$

Solving the system yields 5 solutions:

1. $(u, v) = (-1.000000000000000, 2.139260689980808)$

2. $(u, v) = (0.856124222290431, -4.278918663389620)$

3. $(u, v) = (855836625.917668700218201, -1.333333333333333)$

4. $(u, v) = (-855836631.822430610656738, -1.333333333333333)$

5. $(u, v) = (-1.000000000000000, -2.901165451885570)$

None lie in the feasible region, so no solution.

For pattern $(3, 1)$, problem (Q) is to find (u, v_1, v_2) satisfying:

$$\begin{cases} f_4(u, v_1) = 0 & (1) \\ f_2(u, v_2) = 0 & (2) \\ u^2 = f_3(u, v_1) \cdot f_1(u, v_2) & (3) \\ 0.7 \le u \le \frac{49}{9} \\ 1 \le v_1, v_2 \le 1 + \frac{3}{7}u - \frac{27}{343}u^2 \\ f_1(u, v_1), f_2(u, v_1), f_3(u, v_1), f_1(u, v_2) > 0. \end{cases} \tag{Q}$$

Solving the system yields $(u, v_1, v_2) = (\frac{\sqrt{21}}{3}, \frac{3\sqrt{21}-28}{21}, 1)$, which lies outside the feasible region, so no solution.

For pattern $(2, 2)$, problem (Q) is to find (u, v) satisfying:

$$\begin{cases} f_3(u, v) = 0 & (1) \\ u = f_2(u, v) & (2) \\ 0.7 \le u \le \frac{49}{9} \\ 1 \le v \le 1 + \frac{3}{7}u - \frac{27}{343}u^2 \\ f_1(u, v), f_2(u, v) > 0. \end{cases} \tag{Q}$$

Solving the system yields $(u, v) \approx (1.53, 1.65)$, which lies outside the feasible region, so no solution.

For pattern $(1, 1, 1)$, by the AM-GM inequality, it is easy to see that the minimizer is of the form $(a, 0, a, 0, a, 0)$, and the minimum value is $\frac{30}{7}$. Since $\frac{30}{7} < 6$, it is smaller than the trivial case.

In summary, $P(6, 0.3) = \frac{30}{7}$.

The above summarizes the general method for deriving the minimum of the denominator-weighted Shapiro cyclic sum. However, it is evident that the classification and computational effort are substantial. This method can strictly determine the minimum and minimizer of the Shapiro cyclic sum for small n; for large n, the main computational challenges are: (1) classification of patterns for n, (2) computation of the rational functions $f_i(u, v)$, and (3) solving systems of polynomial equations, all of which are quite difficult. In the following, we propose an efficient algorithm for solving the minimum of the denominator-weighted Shapiro cyclic sum.

4. Efficient RBFGS Algorithm for the Minimum of the Denominator-Weighted Shapiro Cyclic Sum

4.1. Problem Transformation

The BFGS algorithm is an efficient method for solving unconstrained optimization problems. For the denominator-weighted Shapiro cyclic sum problem, let $x_i = y_i^2$. Then $y_i \in \mathbb{R}$ automatically ensures $x_i \ge 0$, transforming the problem into an unconstrained optimization problem in y_i. Implementing the BFGS algorithm for this problem reveals that either the range of feasible initial points is extremely narrow, making it difficult to find counterexamples or to exhibit global properties, or that when the initial point range is slightly expanded, the algorithm quickly diverges to the maximum or minimum representable by computer floating-point arithmetic, yielding poor performance.

Furthermore, since the objective function is homogeneous of degree zero, we can add a normalization condition $\sum_{i=1}^n y_i^2 = 1$, i.e.,

$$\min_{\mathbf{y} \in \mathbb{S}^{n-1}} f(\mathbf{y}) = \sum_{i=1}^{n} \frac{y_i^2}{p y_{i+1}^2 + q y_{i+2}^2},$$

where $\mathbb{S}^{n-1} = \{\mathbf{y} \in \mathbb{R}^n \mid \|\mathbf{y}\|_2 = 1\}$ is the unit sphere, a compact Riemannian manifold.

Definition 4.1. *The unit sphere $\mathbb{S}^{n-1}$ is a Riemannian manifold embedded in $\mathbb{R}^n$ with the following geometric structure:*

- ***Tangent space**: The tangent space at a point $\mathbf{y} \in \mathbb{S}^{n-1}$ is*

$$T_{\mathbf{y}}\mathbb{S}^{n-1} = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{y}^T\mathbf{v} = 0\}.$$

- ***Projection map**: The projection mapping a Euclidean vector onto the tangent space is*

$$P_{\mathbf{y}}(\mathbf{v}) = \mathbf{v} - (\mathbf{y}^T\mathbf{v})\mathbf{y}.$$

- ***Riemannian gradient**: The Euclidean gradient of the objective function is $\nabla f(\mathbf{y})$, and its Riemannian gradient is the projection:*

$$grad f(\mathbf{y}) = P_{\mathbf{y}}(\nabla f(\mathbf{y})).$$

To implement the BFGS algorithm on the manifold, the following concepts are needed:

Definition 4.2. *A retraction $R_{\mathbf{y}} : T_{\mathbf{y}}\mathbb{S}^{n-1} \to \mathbb{S}^{n-1}$ maps a tangent vector back to the manifold, satisfying:*

1. $R_{\mathbf{y}}(\mathbf{0}) = \mathbf{y}$,
2. $\frac{d}{dt}R_{\mathbf{y}}(t\boldsymbol{\eta})|_{t=0} = \boldsymbol{\eta}$.

For the unit sphere, the normalization retraction is used:

$$R_{\mathbf{y}}(\boldsymbol{\eta}) = \frac{\mathbf{y} + \boldsymbol{\eta}}{\|\mathbf{y} + \boldsymbol{\eta}\|}.$$

Definition 4.3 ([9])**.** *A vector transport $\mathcal{T}_{\mathbf{y}\to\mathbf{z}} : T_{\mathbf{y}}\mathbb{S}^{n-1} \to T_{\mathbf{z}}\mathbb{S}^{n-1}$ maps tangent vectors between different tangent spaces, requiring linearity and preservation of inner products. Common approaches include projection transport and parallel transport. This paper adopts parallel transport:*

$$\mathcal{T}_{\boldsymbol{\eta}}(\boldsymbol{\xi}) = \mathcal{T}_{\mathbf{y}\to\mathbf{z}} = \boldsymbol{\xi} - \frac{(\mathbf{y} + \mathbf{z})^T\boldsymbol{\xi}}{1 + \mathbf{y}^T\mathbf{z}}(\mathbf{y} + \mathbf{z}),$$

where $\mathbf{z} = R_{\mathbf{y}}(\boldsymbol{\eta})$.

4.2. Riemannian BFGS (RBFGS) Algorithm Procedure

(summarized from [11][7])

1. Initialization: Choose $\mathbf{y}_0 \in \mathbb{S}^{n-1}$, set $H_0 = I_n$.
2. For $k = 0, 1, 2, \ldots$, do:
   (a) Compute the Riemannian gradient $\mathbf{g}_k = \text{grad} f(\mathbf{y}_k)$. If the norm of the Riemannian gradient is less than a preset tolerance or the maximum number of iterations is exceeded, output $\mathbf{y}_k$ and terminate.
   (b) Compute the search direction $\boldsymbol{\eta}_k = -H_k\mathbf{g}_k$.
   (c) Determine the step size α_k via a Wolfe line search.
   (d) Update the point: $\mathbf{y}_{k+1} = R_{\mathbf{y}_k}(\alpha_k\boldsymbol{\eta}_k)$.
   (e) Compute the transported vectors:

$$\mathbf{s}_k = \mathcal{T}_{\alpha_k\boldsymbol{\eta}_k}(\alpha_k\boldsymbol{\eta}_k), \quad \mathbf{y}_k^{\text{grad}} = \text{grad} f(\mathbf{y}_{k+1}) - \mathcal{T}_{\alpha_k\boldsymbol{\eta}_k}(\text{grad} f(\mathbf{y}_k)).$$

   (f) Update the inverse Hessian approximation using the BFGS formula:

$$H_{k+1} = \left(I - \frac{\mathbf{s}_k(\mathbf{y}_k^{\text{grad}})^T}{\langle \mathbf{s}_k, \mathbf{y}_k^{\text{grad}} \rangle}\right) H_k \left(I - \frac{\mathbf{y}_k^{\text{grad}}\mathbf{s}_k^T}{\langle \mathbf{s}_k, \mathbf{y}_k^{\text{grad}} \rangle}\right) + \frac{\mathbf{s}_k\mathbf{s}_k^T}{\langle \mathbf{s}_k, \mathbf{y}_k^{\text{grad}} \rangle}.$$

4.3. Numerical Experiments and Algorithm Enhancements

Oscillatory Solution Detection and Restart Mechanism

Numerical experiments reveal that for even n, the solution often converges to an oscillatory form $(a, b, a, b, \ldots)$, at which the gradient is zero. Referring to Theorem 3.12, this is likely merely a local stationary point. Moreover, several important counterexamples to the classical Shapiro inequality are near this $(a, b, a, b, \ldots)$ pattern. Therefore, a detection mechanism is added:

$$\text{is_alternating}(\mathbf{y}) = \begin{cases} \text{True}, & \text{std}(y_{\text{odd}}) < \epsilon \text{ and } \text{std}(y_{\text{even}}) < \epsilon \\ \text{False}, & \text{otherwise} \end{cases}$$

When an oscillatory solution is detected with $f(\mathbf{y}) \geq n$, a random perturbation is applied to the current solution:

$$\mathbf{y}_{\text{perturbed}} = \frac{\mathbf{y} + \delta\mathbf{r}}{\|\mathbf{y} + \delta\mathbf{r}\|}, \quad \mathbf{r} \sim \mathcal{N}(0, I_n),$$

and the optimization is restarted from the perturbed point, repeated at most 10 times (or a specified number).

Maintaining the Curvature Condition

From the derivation in Lemma 2.4, in Euclidean space, if the initial H_0 is positive definite and the step size satisfies the curvature condition

$$\nabla f(x^{(k)} + \lambda d^{(k)})^T d^{(k)} \geq c_2 \nabla f(x^{(k)})^T d^{(k)},$$

then the curvature condition and the positive definiteness of H_k hold for all iterations.

On the unit sphere, it often occurs that after several iterations the curvature condition is no longer satisfied. If the iteration continues, the positive definiteness of H_k cannot be maintained, and the function value increases significantly, deviating from the objective. Therefore, a curvature monitoring mechanism is added: if no step size satisfying the Wolfe line search conditions is found, the step size is determined via an Armijo line search, and H_k is reset to I_n.

Additionally, the algorithm incorporates other measures to ensure safe computation: (1) safe norm and safe division to avoid numerical overflow, and (2) gradient clipping to control update magnitudes.

Convergence Analysis

Based on the above content, write a Python program [16] to conduct numerical experiments. The algorithm exhibits good convergence across problems of various scales:

- The gradient norm converges to zero.
- The objective function value of Shapiro cyclic sum converges stably within each run.
- For the four sets of critical values of the classical Shapiro inequality, namely $p = 0.5$, $n = 12, 14$ and $n = 23, 25$, the algorithm performs well, finding counterexamples for $n = 14, 25$ within 10 perturbations.

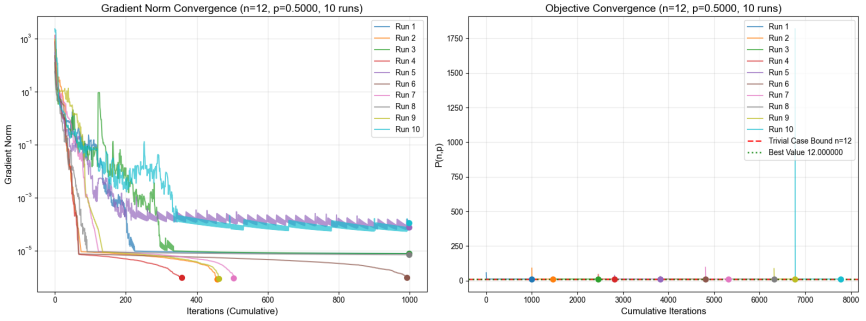


Figure 2: Convergence of Shapiro Cyclic Sum and Gradient Norm ($n = 12, p = 0.5$)

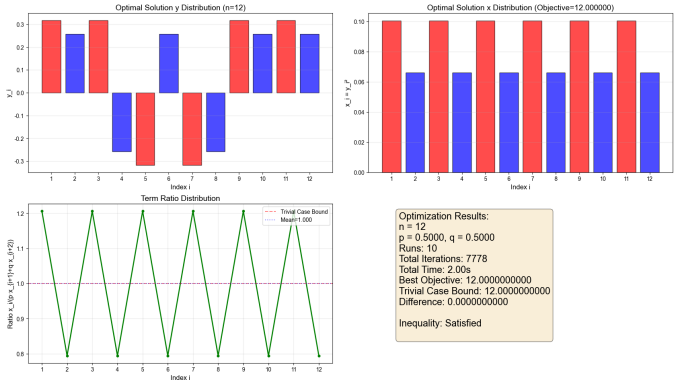


Figure 3: Optimal Solution Distribution ($n = 12, p = 0.5$)

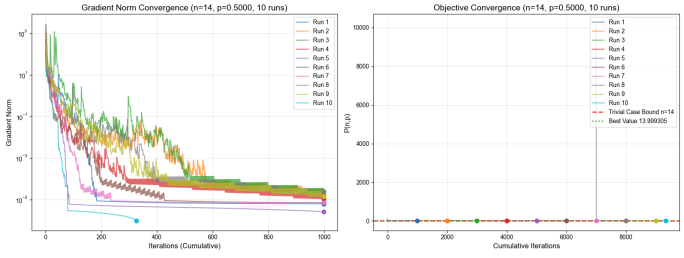


Figure 4: Convergence of Shapiro Cyclic Sum and Gradient Norm ($n = 14, p = 0.5$)

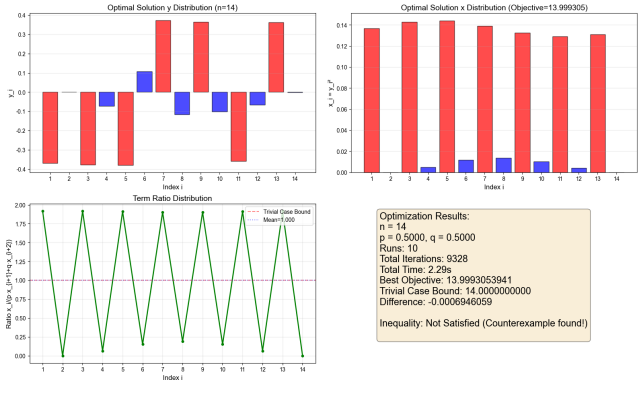


Figure 5: Optimal Solution Distribution ($n = 14, p = 0.5$)

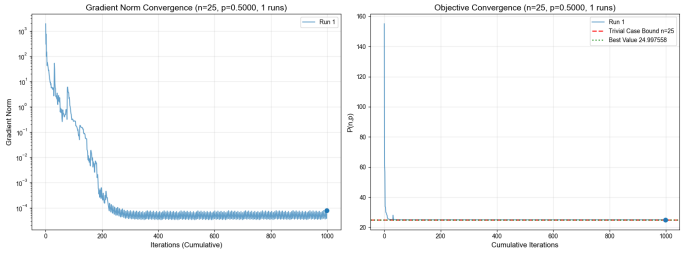


Figure 6: Convergence of Shapiro Cyclic Sum and Gradient Norm ($n = 25, p = 0.5$)

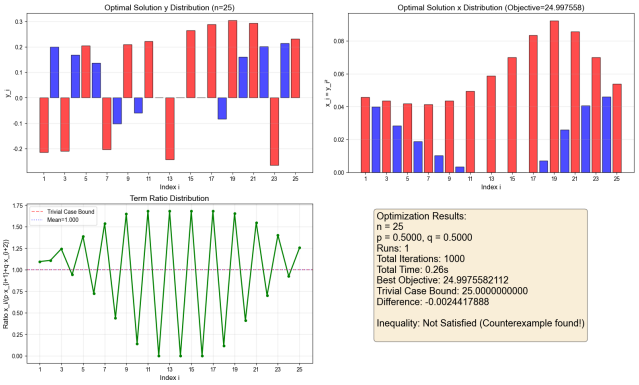


Figure 7: Optimal Solution Distribution ($n = 25, p = 0.5$)

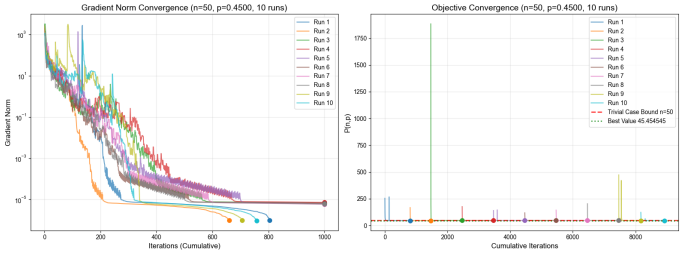


Figure 8: Convergence of Shapiro Cyclic Sum and Gradient Norm ($n = 50, p = 0.45$)

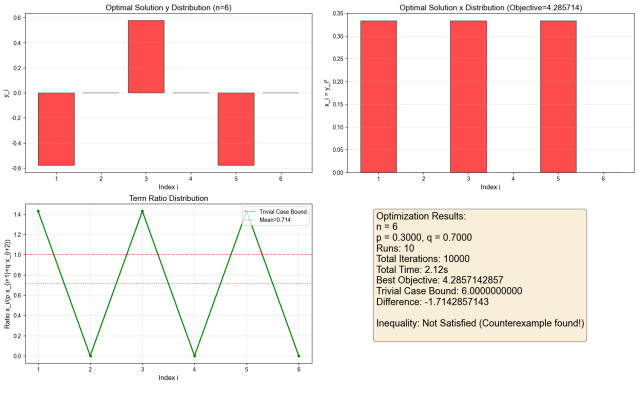


Figure 9: Optimal Solution Distribution ($n = 6, p = 0.3$)

5. Acknowledgments

I would like to express my sincerest gratitude to my instructor in the Mathematical Analysis course and the supervisor of this research project, Professor Shengliang Pan. From the topic selection and literature review to the finalization of the paper, every step was accomplished under Professor Pan's meticulous guidance. It is a great fortune to have had the opportunity to complete this research under his supervision!

I am also grateful to the School of Mathematical Sciences, Tongji University for providing a rich academic atmosphere. This work was supported by the Tongji University Student Innovation Training Program (SITP) under No. X2025634. I hereby express my sincere thanks.

References

[1] T. Ando. A new proof of shapiro inequality. *Mathematical Inequalities & Applications*, 16(3):611–632, 2013. doi: 10.7153/mia-16-46.

[2] P. J. Bushell. Shapiro's cyclic sum. *Bulletin of the London Mathematical Society*, 26:564–574, 1994. doi: 10.1112/blms/26.6.564.

[3] P. J. Bushell and J. B. McLeod. Shapiro's cyclic inequality for even n. *Journal of Inequalities and Applications*, 7(3):331–348, 2002. doi: 10.1080/10255830290030426.

[4] B. L. Chen. *Optimization Theory and Algorithms*. Tsinghua University Press, Beijing, 2nd ed. edition, 2005. in Chinese.

[5] P. N. Choudhury and K. C. Sivakumar. Nesbitt and shapiro cyclic sum inequalities for positive definite matrices. *Advances in Operator Theory*, 7(1):7, 2022. doi: 10.1007/s43036-021-00171-0.

[6] N. Y. Deng. *Unconstrained Optimization Computational Methods*. Science Press, Beijing, 1982. in Chinese.

[7] W. Huang, K. A. Gallivan, and P.-A. Absil. A broyden class of quasi-newton methods for riemannian optimization. *SIAM Journal on Optimization*, 25(3):1660–1685, 2015. doi: 10.1137/140955483.

[8] M. T. Nguyen and Q. T. Le. On an extension of shapiro's cyclic inequality. *Journal of Inequalities and Applications*, page Article ID 491576, 2009. doi: 10.1155/2009/491576.

[9] C. Qi. *Numerical optimization methods on Riemannian manifolds*. PhD thesis, The Florida State University, Tallahassee, 2011.

[10] S. Radulescu, M. Radulescu, J. L. Diaz-Barrero, and P. Alexandrescu. Two families of cyclic inequalities. *Mathematical Inequalities & Applications*, 15(1):199–209, 2012.

[11] W. Ring and B. Wirth. Optimization methods on riemannian manifolds and their application to shape space. *SIAM Journal on Optimization*, 22(2):596–627, 2012. doi: 10.1137/11082885X.

[12] J. M. Steele. *The Cauchy-Schwarz Master Class: An Introduction to the Art of Mathematical Inequalities*. Cambridge University Press, Cambridge, 2004.

[13] T. Tanriverdi. Reformulation of shapiro's inequality. *International Mathematical Forum*, 7(43):2125–2130, 2012.

[14] B. A. Troesch. On shapiro cyclic inequality for n = 13. *Mathematics of Computation*, 45(171):199–207, 1985. doi: 10.2307/2008058.

[15] B. A. Troesch. The validity of shapiro cyclic inequality. *Mathematics of Computation*, 53(188):667–664, 1989. doi: 10.2307/2008728.

[16] A. Wang. Code for the minimum of the denominator-weighted shapiro cyclic sum via rbfgs, 2026.